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[22BSMA41]

FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

OPTIMIZATION TECHNIQUES

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**05 X 02=10**		
1.	**a.**	Define Infeasible and Unbounded Solution.	L-2	CO1	**02**
	b.	Change the following unbalanced transportation problem into balanced one.	L-2	CO1	**02**

	X	Y	Z	Availability
A	4	3	2	10
B	5	6	1	8
C	6	4	3	5
D	3	5	4	6
Demand	7	12	4	

	c.	Define two-person zero-sum game with suitable example.	L-2	CO1	**02**
	d.	Define total float and free float.	L-2	CO1	**02**
	e.	Write the Kendall's notation for representing queueing models.	L-2	CO1	**02**
		PART-B	**06 X 05=30**		
2.	**a.**	Explain the phases of Operations research Problem.	L-2	CO2	**05**
		OR			
	b.	A manufacturer produces two types of models M1 and M2. Each model of the type M1 requires 4 hours of grinding and 2 hours of polishing; whereas each model of the type M2 requires 2 hours of grinding and 5 hours of polishing. The manufacturer has 2 grinders and 3 polishers. Each grinder works 40 hours a week and each polisher work for 60 hours a week. Profit on M1 model is Rs. 3.00 and on model M2 is Rs. 4.00. Whatever is produced in a week is sold in the market. Formulate how should the manufacturer allocate his production capacity to the two types of models so that he may make the maximum profit in a week.	L-2	CO2	**05**

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3.	**a.**	Obtain initial basic feasible solution to the following transportation problem using North-West corner method.	L-3	CO3	**05**

	X1	X2	X3	Supply
A	7	6	9	20
B	5	7	3	28
C	4	5	8	17
Demand	21	25	19	

OR

	b.	Obtain initial basic feasible solution to the following transportation problem using matrix minima method.	L-3	CO3	**05**

	X1	X2	X3	X4	Supply
A	1	2	3	4	6
B	4	3	2	0	8
C	0	2	2	1	10
Required	4	6	8	6	

4.	**a.**	Solve the game whose pay off matrix is given below.	L-3	CO4	**05**

		Player A				
Player B		B1	B2	B3	B4	B5
	A1	-2	0	0	5	3
	A2	3	2	1	2	2
	A3	-4	-3	0	-2	6
	A4	5	3	-4	2	-6

OR

	b.	Solve the following game and determine its value.	L-3	CO4	**05**

PLAYER A	PLAYER B		
		B1	B2
	A1	1	-1/2
	A2	-1/2	0

5.	**a.**	Explain the rules of network construction.	L-2	CO5	**05**

OR

	b.	Construct a network for the project whose activities and precedence	L-2	CO5	**05**

relationships are as given below.

Activities	A	B	C	D	E	F	G	H	I
Predecessor	-	A	A	-	D	B, C, E	F	D	G, H

6.	**a.**	Draw a schematic representation of queueing system and also explain arrival pattern and the service mechanism.	L-2	CO6	**05**
		OR			
	b.	People arrive at a theatre ticket booth in Poisson distributed arrival rate of 25 per hour. Service time is constant at 2 minutes. Find **i.** The mean number in the waiting line **ii.** The mean waiting time **iii.** The utilization factors.	L-3	CO6	**05**
7.	**a.**	Determine initial basic feasible solution for the following transportation problem using VAM.	L-3	CO3	**05**

	X1	X2	X3	X4	Supply
A	11	13	17	14	250
B	16	18	14	10	300
C	21	24	13	10	400
Demand	200	225	275	250	

		OR			
	b.	Convert the following game as linear programming problem.	L-3	CO4	**05**

PLAYER A	PLAYER B		
	B1	B2	B3
A1	3	-4	2
A2	1	-3	-7
A3	-2	4	7

		PART-C	**06 X 10=60**		
8.	**a.**	Solve the following problem graphically	L-3	CO2	**10**

$$Maximize Z = 5x_1 + 7x_2$$
$$Subject to$$
$$x_1 + x_2 \leq 4$$
$$3x_1 + 8x_2 \leq 24$$
$$10x_1 + 7x_2 \leq 35$$
$$x_1, x_2 \geq 0$$

Explain the graphical Solution.

		OR			

	b.	A company makes two kinds of leather belts, belt A and belt B. Belt A is a high-quality belt and belt B is of lower quality. The respective profits are Rs 4 and Rs 3 per belt. The production of each of type A requires twice as much time as a belt of type B, and if all belts were of type B, the company could make 1,000 belts per day. The supply of leather is sufficient for only 800 belts per day (both A and B combined). Belt A requires a fancy buckle and only 400 of these are available per day. There are only 700 buckles a day available for belt B. What should be the daily production of each type of belt? Formulate this problem as an LP model and solve it using the simplex method.	L-3	CO2	**10**
9.	**a.**	Solv the following transportation problem with initial basic feasible solution using VAM.	L-3	CO3	**10**

	D1	D2	D3	D4	Supply
S1	6	1	9	3	70
S2	11	5	2	8	55
S3	10	12	4	7	70
Demand	85	35	50	45	

OR

	b.	A company has 4 machines to do 3 jobs. Each job can be assigned to only one machine. The cost of each job on each machine is given below. Determine the job assignment that will minimize the total cost.	L-3	CO3	**10**

	W	X	Y	Z
A	18	24	28	32
B	8	13	17	18
C	10	15	19	22

10.	**a.**	Using the principle of dominance, solve the following game.	L-3	CO4	**10**

		PLAYER B				
		B1	B2	B3	B4	B5
PLAYER A	A1	4	6	5	10	6
	A2	7	8	5	9	10
	A3	8	9	11	10	9
	A4	6	4	10	6	4

OR

	b.	Solve the following game by graphical method.	L-2	CO4	**10**

		PLAYER B		
		B1	B2	B3
PLAYER A	A1	3	-1	0
	A2	2	1	-1

11.	**a.**	A project schedule has the following characteristics.	L-3	CO5	**10**

Activity	1-2	1-3	2-4	3-4	3-5	4-9	5-6	5-7	6-8	7-8	8-10	9-10
Time	4	1	1	1	6	5	4	8	1	2	5	7

From the above information, you are required to:

i. Construct a network diagram.

ii. Compute the earliest event time and latest event time.

iii. Determine the critical path and total project duration.

iv. Compute total and free float for each activity.

OR

	b.	A small project is composed of seven activities, whose time estimates are listed in the table as follows	L-3	CO5	**10**

Job	1-2	1-3	1-4	2-5	3-5	4-6	5-6
a (Optimistic)	1	1	2	1	2	2	3
m(Most likely)	1	4	2	1	5	5	6
b(Pessimistic)	7	7	8	1	14	8	15

i. Draw the project network

ii. Find the expected duration and variance of each activity

iii. Calculate the variance and standard deviation of project length

iv. What is the probability that the project will be completed

- **i.** 4 weeks earlier than the expected,
- **ii.** Not more than 4 weeks later than expected,
- **iii.** If the project's due date is 19 weeks, what is the probability of meeting the due date?

12.	**a.**	The arrival rate of customers at the single window-booking counter of a two-wheeler agency follows Poisson distribution and the service tine follows exponential (negative) distribution and hence, the service rate follows Poisson distribution. The arrival rate and the service rate are 25 customers per hour and 35 customers per hour, respectively. Find the following:	L-3	CO6	**10**

		i. Utilization of the booking clerk. **ii.** Average number of waiting customers in the queue. **iii.** Average number of waiting customers in the system. **iv.** Average waiting time per customer in the queue. **v.** Average waiting time per customer in the system.			
		OR			
	b.	Strong is a dentist who schedules all her patients for 30 minutes appointments some of the patients take more or less than 30 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and the time actually needed to complete the work:	L-3	CO6	**10**

Category	Time required (in minutes)	Number of patients
Filling	45	40
Crown	60	15
Cleaning	15	15
Extracting	45	10
Check up	15	20

Simulate the dentist's clinic for four hours and find out the average waiting time for the patients as well as the idleness of the doctor. Assume that all the patents show up at the clinic at exactly their scheduled, arrival time starting at 8:00 am. Use the following random numbers for handling above problem.40, 82, 11, 34, 25, 66, 17 and 79.

13.	**a.**	Solve the following LPP using Two-Phase method $MinZ = 12x + 18y + 15z$ $Subjectto$ $4x + 8y + 6z \geq 64$ $3x + 6y + 12z \geq 96$ $x, y, z \geq 0$	L-3	CO2	**10**
		OR			
	b.	Solve the following game by graphical method.	L-3	CO4	**10**

PLAYER A	PLAYER B	
	B1	B2
A1	-6	7
A2	4	-5
A3	-1	-2
A4	-2	5
A5	7	6